

Executive Summary

Waterloo Rocketry is a student design team at the University of Waterloo that builds and flies rockets every year at the Launch Canada Challenge. The vehicle flown in 2024, *Borealis*, presented numerous interesting systems and integration challenges throughout design, testing and flight. The intent of this report is to describe and characterize the integration challenges faced throughout the entire life cycle of *Borealis* and provide some key steps for improvement and continued growth that will be critical as the team takes on more interesting and complex engineering challenges. This report was written by the 2024 integration lead, and focuses primarily on the system-interface level, rather than detailing any particular system.

The report opens with a discussion of what technical and non-technical attributes an integration lead must have. In particular, an integration lead should have foresight, excellent technical communication, and a clear sense of priorities. An integration lead should also have the vision required to understand the ultimate direction of the vehicle and direct their teammates towards that final goal.

Then, there is some discussion on requirements, how to set requirements, and how to ensure that all the requirements are well thought-through. This involved pushing the requirements definition down to the lowest possible level, the subsystem leads, then mixing and coordinating those requirements as they interacted with each other. Though these requirements were rarely referenced after the definition, the integration lead developing a clear picture of what to expect from the upcoming rocket was an immensely useful exercise. This is one of the weakest phases of Waterloo Rocketry's systems design engineering process, and improvements to the future process are expanded upon.

Preliminary design of the rocket was a large negotiation between project leads. It took the form of meetings, sketches, and debates being shared around until a final architecture was chosen. Many of the final layout decisions were obvious, forced by the inherent physics of rocket flight. Some of the decisions were made solely on the basis of flight heritage, which, for the team's purposes, is a good justification. Ultimately, all of the layout and section details fell into place, and work began on more detailed design.

Detailed design took place in Solidworks using skeleton CAD, an industry standard practice that defines complex CAD assemblies the way that software engineers build computer systems. It is an interface-first sketching system that is used to define all relevant parameters. If done properly, then any changes to the CAD will cascade down into the child features, leaving all interfaces intact.

There were 3 major integration tests for *Borealis*, rocket assembly day, wet dress rehearsal, and competition simulator. Rocket assembly day involved putting the entire vehicle together, ensuring that the

airframe sections agreed with each other, and that there were no unpleasant integration surprises in the upper half of the rocket. Wet dress rehearsal involved assembling the bottom half of the rocket, filling it with simulated propellants, and ensuring that the rocket fluid and electrical systems could operate nominally. Finally, competition simulator involved taking the rocket out to a field, doing a full assembly of the vehicle, then working through the full nominal operation procedure. This was a test of the available procedures, timelines, and readiness of the vehicle. All tests had significant issues but passed reasonably well.

Finally, there is discussion on the assembly, operations, and flight of *Borealis*. Risk analysis was a notably weak process in the 2024 design cycle, and the team would have benefited from a much more heavy-handed risk analysis process. Several recommendations are made to future integration leads.

Introduction

Waterloo Rocketry is a student design team at the University of Waterloo, who's primary objective is to teach engineering students the skills required to be effective engineers. The primary medium through which this objective is achieved is through the design, build, and test of a sounding rocket flown each year at the Launch Canada Challenge in Timmins, Ontario. The sounding rocket is composed of a number of different systems (propulsion, aerostucture, avionics, etc.) which all must coordinate and agree with each other to create a unified vehicle. This is the responsibility of the integration lead, which coordinates all systems to build a single coherent vehicle.

The coordination of all of these systems by the integration lead is a difficult task with many dynamic parts. The objective of this report is to review the integration efforts from 2024 and to provide support and advice to any future integration leads who attempt similarly difficult things.

This work term report will outline the steps, practices, procedures, and lessons learned from the integration lead's efforts in building *Borealis*, such that future integration leads can learn from this experience and avoid making similar mistakes.

Problem Definition

There exists a sounding rocket that requires coordination between the various systems to build a single coherent vehicle. Requirements must be made coherent and internally consistent, common interfaces must be defined and organized, and vehicle-level decisions must be made and communicated to the rest of the team members working on the development of the vehicle.

Informality of Process

Waterloo Rocketry is composed entirely of volunteer students from across the University of Waterloo. This means they arrive with a variety of baselines in engineering skills, professional experience, and knowledge of industry practices. This means that the systems design process is not always as effectively laid out as it would be in a professional environment, with full time staff and explicit responsibility. Additionally, the team often builds on historical experience, design of past vehicles, and technical heritage. This means not every requirement is questioned and not every system is optimized. This is required to make the vehicle meet the competition deadlines and is something the team benefits greatly from. However, it also means that knowledge of vehicles must be passed down effectively between team members such that past experience does not get forgotten.

It is also important to note that due to the amateur nature of the project, complex analysis is limited and wide performance margins are left. Detailed analysis of components is quite rare, largely due to lack of time and resources. Instead, the team prefers to build vehicles that are more than capable of supporting the projects team members would like to fly and focus less on optimization and mass requirements. It is a key instinct of an integration lead to develop an understanding of when to continue to optimize and push for higher performance, and when to accept an imperfect design. The integration lead should also keep in mind that the highest priority of the vehicle is flight and prefer systems that will support flight under any conditions rather than highly optimized performance.

Vehicle Design Preparation

Before design of a particular vehicle begins, the integration lead should give some thought ahead of time to the following things.

- Who should be responsible for what level of interface. Historically, the inter-system interfaces have been the domain of the system leads, with the intra-system interfaces being the responsibility of the integration lead. However, this may change for particular technical domains, if there are (for example) particularly new system leads or particularly generalist system leads
- Where does accountability fall for each encountered issue. The integration lead is officially responsible for the entirety of the vehicle, so it is critical to determine who is reliable and trustworthy, who requires excess support, and who can be brought in to help in the case of specific issues.

There are also some documents and books that are useful for the integration lead to read and understand. *Managing Complex Technical Projects: A Systems Engineering Approach* by R. Ian Faulconbridge and Michael J. Ryan [1] is an excellent textbook on how to organize and work through various principles of systems engineering. Chapters 2-5 are especially useful, though do understand that these principles are for projects much larger than the team's rockets and do not always apply. The *Nasa Systems Engineering Handbook* [2] is largely a second description of the same content of the previous text but also provides some excellent visuals and diagrams to better understand the content. Finally, the paper "What Made Apollo A Success" [3] provides an excellent insight into how the Apollo program was arranged technically and effectively emphasizes the importance of simple interfaces that require minimal interpersonal communication.

Integration leads would also do well to have and maintain some set of technical and non-technical skills. The first is effective and clear communication. Communication of interfaces is complex and

often flight-critical, especially between systems that do not frequently interact. Therefore, the integration lead must be able to clearly state what the interface looks like, ensure that the relevant parties understand that interface, and identify when teammates' communication is unclear and how to resolve that complication. Communication occurrences where both parties leave completely convinced of different ideas push timelines and kill flights. The integration lead would also do well to have both strong vehicle-level knowledge and good systems design principles. They should understand both the general makeup of the vehicle and how various projects fit together, as well as how to do effective requirements definition and risk analysis. Finally, the integration lead must possess immense foresight, identifying technical problems well in advance and coordinating the solutions, ideally before the affected parties have begun work on that system. Remaining weeks or months in advance ahead of the rest of the team allows the integration lead to reduce wasted effort and keep track of project-killing issues long before any steps backwards have to be taken. It is the integration lead's responsibility to see through the fog of vehicle design and assembly, this is a skill they would do well to cultivate.

Finally, the integration lead must keep in mind that the ultimate priority for the flight is to launch. Launching the vehicle and safe recovery is the primary success criteria for the vehicle, and the integration lead is responsible for choosing the path that will bring the vehicle closest to launch without the addition of other complexity. If the relevant decision will make launch more difficult but improve performance, the correct balance on the compromise being presented is typically in favour of launch.

Requirements Definition

Vehicle-level requirements definition were collected as follows

1. The integration lead selected the key domains of requirements to collect (engine, aerostructure, payload, aerodynamics, etc)
2. The leads for those specific field were responsible for laying out the requirements for those sections
3. The integration lead was responsible for describing, clarifying, and coordinating all the overlapping requirements

A more detailed flowchart of the process can be found in Figure 1.

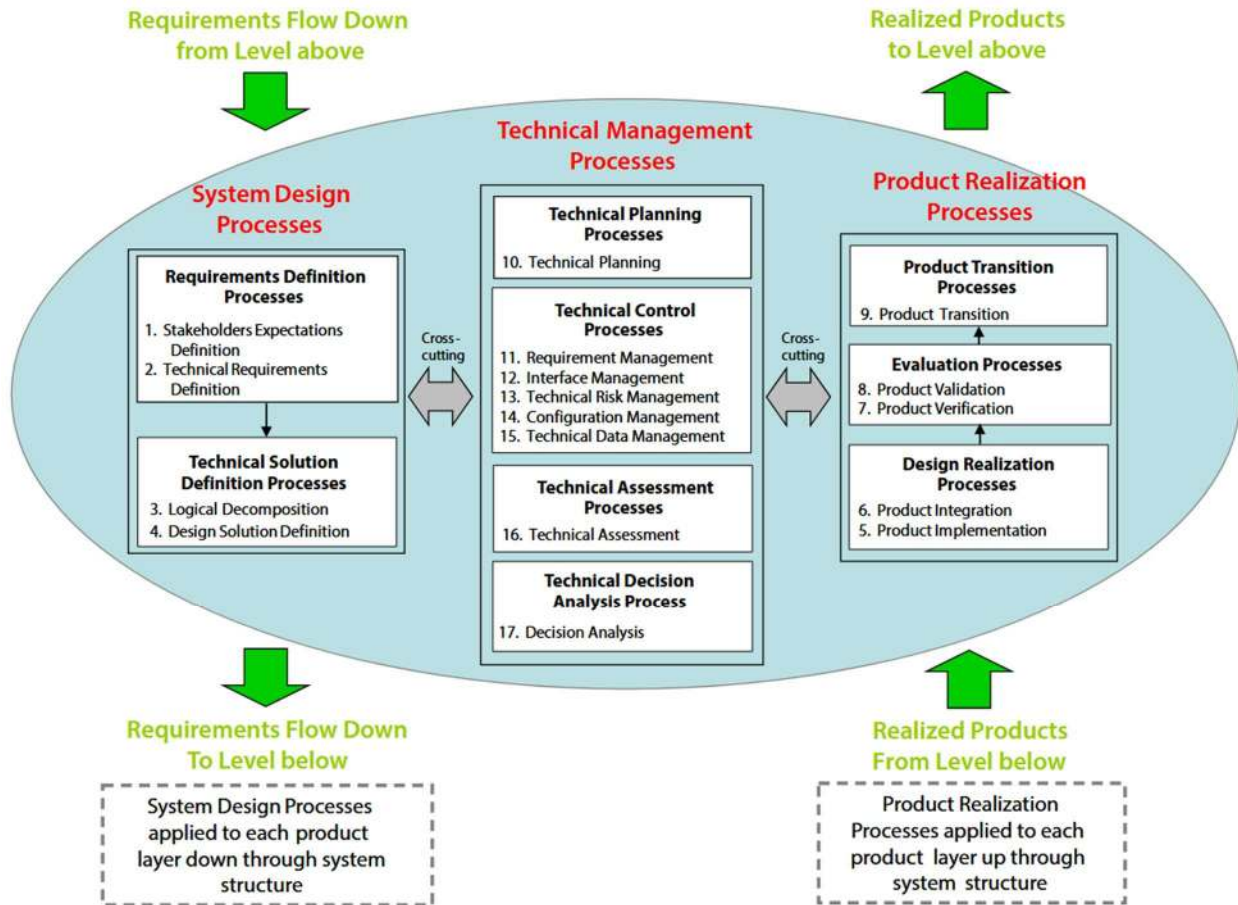


Figure 1: The NASA systems engineering process [1]

For Borealis, this looked like all the system leads collecting relevant requirements in a spreadsheet, and the integration lead combing through that spreadsheet and deciding which requirements were acceptable, which needed further clarification, and which needed to be resolved. The payload lead for *Borealis* had never written requirements before, and so the integration lead supported them throughout that process, guiding them on how to write effective requirements and determine success criteria. The spreadsheet for *Borealis* requirements can be found here¹. Observe that the engine and infrastructure requirements and not well matured as these systems were well understood, this is an example of relying on flight heritage discussed in the section titled **Informality of Process**.

Requirements are also often described directly in project proposal documents, where each of the requirements for a given project are established. Typically, these requirements come from the project owner, with little connection between the project proposal requirements and the higher-level system

¹ https://docs.google.com/spreadsheets/d/1PAj237DSajzMji8_IR-m1vq0bvYTsRZYwTDHQycgdj4/edit?gid=803878771#gid=803878771

requirements found in the spreadsheet mentioned previously. Effective requirements definition in project proposals can be found here². Observe that there is no relationship between the requirements in this document and the requirements in the spreadsheet in the previous section.

Not all systems require explicit requirements definition. Informally, requirements can be largely kept in the project owner's head and disagreements can be resolved through conversation with the relevant parties. Accurate accounting of how requirements have changed is typically unnecessary, so long as all the relevant parties are in agreement. This is a major benefit of working on a student team; the relevant parties are easily accessible and the project is accountable only to the team internally, rather than any external parties who may require more stringent justification. When changing requirements, it is critical that all decisions are made in public spaces with clear attention being called to the change so all relevant parties are made aware. Historically, the team has struggled with requirements changing and not everyone involved being informed, which can have drastic consequences closer to deadlines when it becomes clear that a new requirement was not effectively communicated.

The team historically has not done particularly strict requirements verification at the end of the project, instead preferring owner opinion and integration testing to validate whether something has met the requirements. Typically, the project lead can identify whether a particular system will be effective or not at meeting the requirements without going through each one. However, for systems where the end use case is ambiguous to the designer, more extensive requirements verification is encouraged (i.e. Omnibus being tested in its proper use case rather than with quick, non-representative tests on the developer's personal computer). As the team grows in size and projects grow in complexity, more careful requirements verification may become necessary.

Preliminary Design

Preliminary Design for *Borealis* largely took place through many hand sketches and regular meetings and communication with the other vehicle designers. There were two key fields of vehicle design, the first was initial mass, thrust, and burntime targets to create a target apogee. These decisions were made by the Flight Dynamics lead and will not be discussed in this report. The second area of vehicle design was in the physical layout of the rocket and where various systems would go.

Much of the physical layout of *Borealis* was forced via the physics. The design inherits from previous years' architecture, which, above the engine was quite similar. A visual determination of where various

² <https://docs.google.com/document/d/1FvxAy88NmIJRTLFJIHnnjEJLtoKl4ovmP8m6fwUxZ8/edit?tab=t.0>

systems on the vehicle were to go can be found in Figure 2. The load path of the rocket in various phases of flight is shown in Figure 3. The physical architecture was decided to be as follows:

- Recovery system – must go at extreme end of vehicle due to separating joint, other extreme end of vehicle is engine
- Recovery Electronics – Must be physically close to recovery system
- Cubesat Payload – Most natural position following all other justification
- Airbrakes – Placed as low as possible to minimize impact on vehicle stability
- Engine – Required to be at bottom of vehicle

This architecture was in some sense obvious from the beginning of the design cycle, however it was still a useful exercise to determine it from “first principles” to better understand the requirements and flexibility inherent in each system. All integration leads are encouraged to do this exercise, especially for their first vehicle.

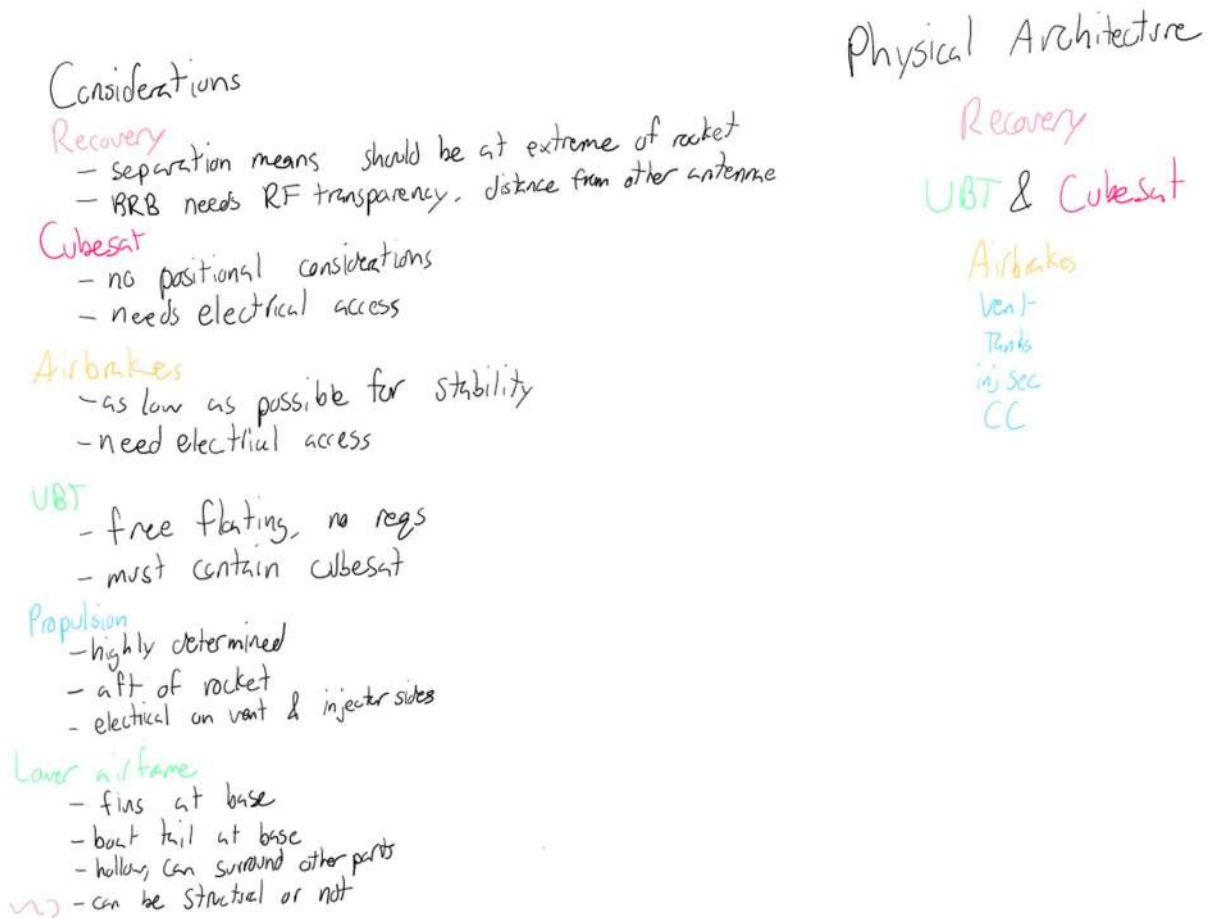


Figure 2: Considerations for Physical System Position on Borealis

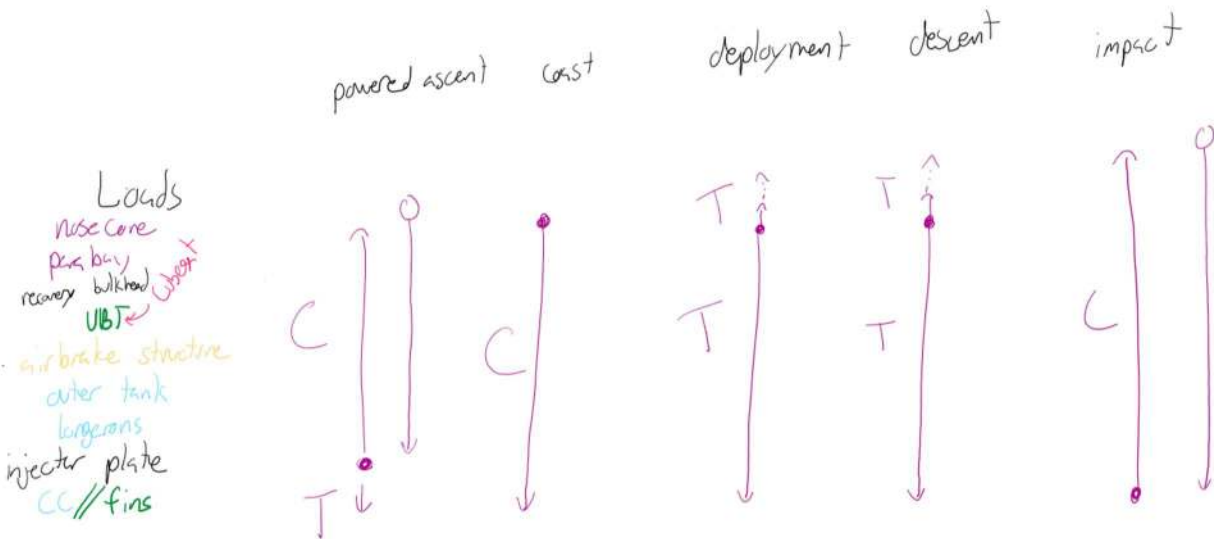


Figure 3: Resulting Load Profile for Decided Physical Architecture

Of the systems that were not obligated by the physics such as the use of CAN or reefed parachutes, one critical factor driving decision making was flight heritage. Many decisions went unquestioned and “it’s what was done last year” was a frequent motivation. Ideally, with every new vehicle, past decisions are interrogated and fully understood. However, timeline and availability pressures of a volunteer team frequently win out and design heritage is used extensively. Integration leads should pay attention to each system on the vehicle and be careful to apply “it’s what was used last year”. It can be an excellent tool, but only in the right circumstances.

For significant changes from previous vehicles, large amounts of time were devoted early on to fully understanding their impacts. This time largely was invested by the flight dynamics lead to investigate aerodynamic effects and the integration lead to investigate overall vehicle architecture. These large-scale changes are discussed below.

It was also useful to anticipate integration problems in advance. It was known well ahead of time that integration of the inter-tank was going to be very challenging. Much effort was spent ahead of time removing components from the injector section, planning its layout, and understanding collectively that this was going to be a difficult problem that would require a lot of human hours. The team also correctly anticipated that unknown issues (like airbrakes interference with fins) would become difficult problems, even if the difficulty of the problem was not well known, so work on these problems began well in

advance of when it was required. For known issues with flight heritage, the team relied on that flight heritage, for unknown issues, the team was correctly paranoid.

Critical Decisions

Several key decisions were made on the vehicle early on that determined how the rest of the vehicle grew around it. These were often a combination of physics and engineering decisions.

First, it was decided early on that *Borealis* would be the team's first attempt at a bi-propellant liquid rocket. This dramatically changes the tank architecture, flight characteristics, and mass distribution, as well as having some downstream effects on things like fin sizing. This decision was made largely to create interesting technical challenges on the team, though the technical benefits (like ability to perform longer burns and higher propellant efficiency) are also enjoyed.

One area that was not well considered was the electrical architecture of the vehicle. *Borealis* had electronics in four different sections of the vehicle and three different power sources, with a number of complex connections between the various sections. This caused enormous difficulty in assembly and test, resulting in long harness build times and damaged wiring, as well as complex assembly operations and difficult deintegration. This is largely due to the integration lead's lack of familiarity with the electrical systems, and can be resolved by seeking out more input from experts on key decisions.

The decision that absorbed the most amount of resources and time by far was the location of the airbrakes. Airbrakes were a new project for Waterloo Rocketry in 2024, and it was critical that they be well understood and integrated. Additionally, while other systems could rely on flight heritage, the airbrakes had to be understood and characterised from a point of unfamiliarity. Many questions were investigated relating to the airbrakes' impact on the aerodynamic stability on the vehicle. For mechanical layout purposes, it had to be decided where the airbrakes should be positioned. Two key locations were considered early on, between the tanks and the injector section or forward of the tanks. The location in the injector section was much preferable for aerodynamic stability, improving stability rather than harming it. However, the engineering challenges of integrating a complex mechanical system around the packed injector section became clear. For this reason, the team preferred to demonstrate that the impact on stability from airbrakes above the tank was minimal, then move them to immediately above the tanks. While this position did harm stability, it preserved simple interfaces and allowed the airbrakes mechanism to operate largely independently, improving final vehicle integration. It is the opinion of the integration lead that if airbrakes were required to be placed in the injector section, then they would have never flown.

Detailed Design and CAD

After establishing rocket architecture and preliminary design of the vehicle, detailed design of the vehicle could progress. In the best case, design of the different systems can be largely independent and coordination can remain simple. However, some complex interfaces remain.

One key consideration during detailed design is to force interfaces to remain simple at every opportunity. Often there is temptation to make an interface more complex for some small performance gain, but this will penalize the testing and operational cadence of the vehicle for performance gains that are often not necessary. Launch is the top priority.

Since 2023, the team has largely relied on SolidWorks-base CAD to define and establish interfaces between owners. This is done using Skeleton CAD, an industry standard practice which forces all interface definition into sketches, with derived geometry being built on these sketches. This allows the vehicle to exist primarily as a series of malleable sketches that can change length, thickness, or position at will. If implemented correctly, then the derived geometry will update automatically when the interfaces change. Skeleton CAD implementation is quite difficult, and the team is still learning best practices, but it has made the rocket CAD more useable and flexible.

This is also the period at which assembly order should be carefully considered and designed for. Coupler geometry should account for assembly order, battery and other energetic placement should consider final assembly of the vehicle, and harness connection points should be determined. Assembly order does not just include the assembly procedure at competition for flight, but also assembly for every test, wet dress rehearsal, competition practice, and mechanical or electrical checkout. Assembly should be both effective for flight and easy, as no assembly is only done once.

One example of a system that required an interface with the airframe was the novel airbrakes. The integration lead and the mechanical designer of the airbrakes spent several hours in a few different calls sorting out what the airbrakes interface with the neighbouring sections should look like, as well as different configurations for the airbrakes internally. The final solution that was settled on was that shown in Figure 4. This interface was followed throughout detailed design and eventually implemented into the flight vehicle, with some mild adjustments for geometry.

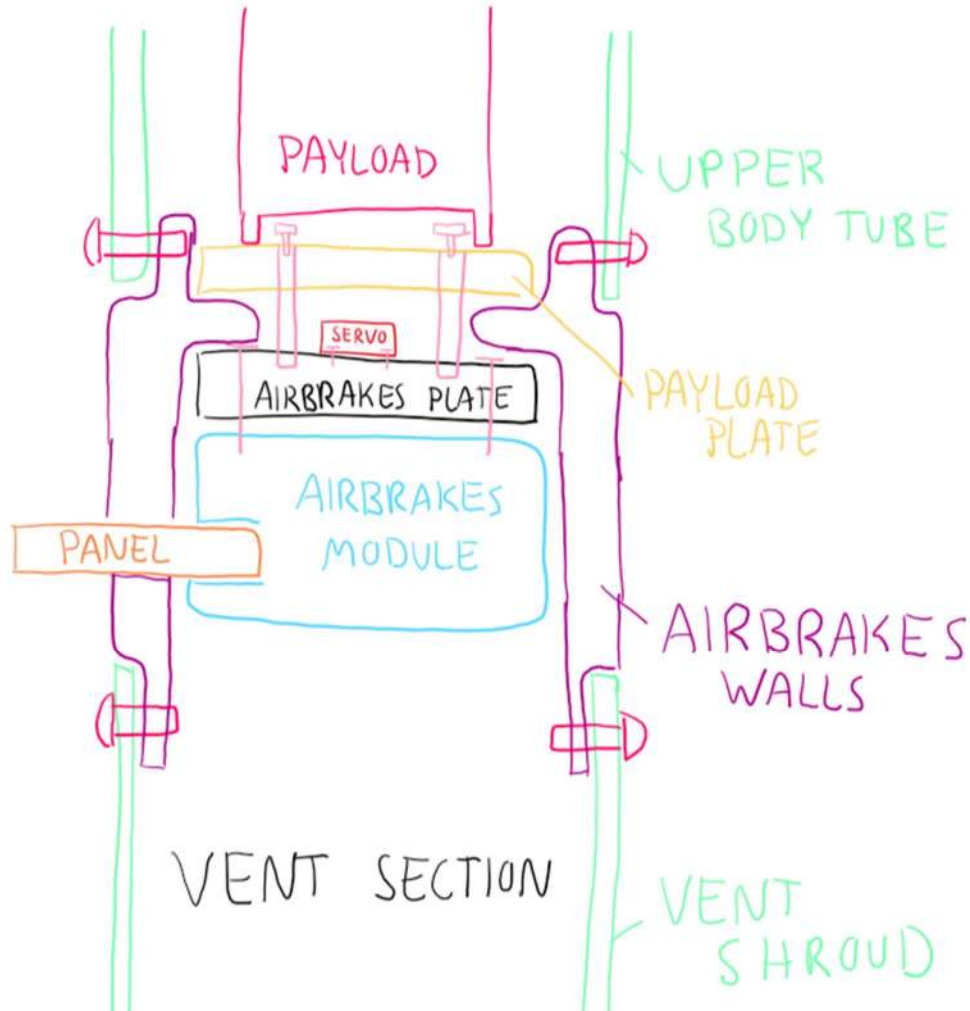


Figure 4: Airbrakes Mechanical Design

Similar to electrical architecture not being considered in preliminary design, harnesses were not well considered in detailed design. Little consideration was given to harness routing, attachments, manufacturing, or timelines. This left harnesses as an afterthought with very little support or schedule for them. This is again attributable to the integration lead not being familiar with electrical requirements and not asking for sufficient support, and it is critical that future integration leads do not make the same mistake.

Design Reviews

One underutilized tool during the development of *Borealis* was design reviews. Typically, systems would be reviewed around the end of detailed design but before manufacturing, and this is

typically far too late to catch any material issues. Design reviews should happen sooner and more frequently, keeping everyone up to speed and allowing relevant stakeholders a chance to understand and comment on the project as it is developed. Ideally, there are at least two design reviews for each major project. The first should occur when the ideation is only sketches and should be primarily a review of requirements and overall project direction. The second should occur when the design is settled, and the project owner is ready to start purchasing materials or otherwise implementing the solution. This should be a review of whether the proposed solution meets the requirements.

Testing

There were three primary mechanical full-scale integration tests undertaken by the team in the development of *Borealis*. The first is Rocket Assembly, the first time all the sections of the rocket fit together and all of the systems are packed inside. This tests the airframe manufacturing, the recovery packing, the assembly order, and frequently sets the manufacturing schedule for the majority of components on the vehicle. The second is Wet Dress Rehearsal, this is an integrated test of the fluid systems and the electronics, taking the vehicle from the beginning of flight operations all the way up to launch, then aborting. Finally, new in 2024, the team attempted to run through a full assembly and operation of *Borealis*, gathering as many hands as possible from the available members and doing a full competition practice run. This event was not as organized or effective as it could have been, but does have promise for future years and vehicles now that it is better understood.

Rocket Assembly took place on June 29th, 2024, 34 days (L-34) before Aug 2nd (the last day before exams, after which team members would be largely unavailable). On this day, the majority of the physical components of the vehicle were assembled, as described above. Rocket Assembly in 2023 was done much later in the design cycle (June 4, L-18) with a much more complete vehicle and with complete procedures. In retrospect, it would have been preferable to do rocket assembly with a complete payload and fabricated harnesses, but the test was still useful, especially as a clear deadline by which all mechanical components on the vehicle were required to be complete. Assembly procedures are also highly encouraged for this test.

Wet Dress Rehearsals (WDR) began on July 6th (L-27) and ended on July 27 (L-6) (2023: WDR1 occurred on L-38, WDRs end L-15). Around 10 distinct operations were carried out in total over 3 weekends and 2 weekday evenings. In each of these tests, the team characterized as many different operational environments and aborts as possible, testing all electronics, sensor failures, abort modes, communication over antenna and ground umbilical, etc. A number of problems were discovered during

WDR, which could then be resolved for the next WDR. One key mistake that was made is that the test cadence was often too fast to resolve all the issues discovered during previous WDRs. This would leave problems unsolved that would reappear in future tests. In future, tests should be delayed to give time for all issues to be resolved, as this also gives team members time to rest and recover before beginning a new test.

The final integration test undertaken by Waterloo Rocketry in 2024 was Competition Simulator (July 27, L-6), a team first. This was intended to be an end-to-end test of competition operations, going through operating procedures, launch pad setup, vehicle assembly, and a full fill and abort of the vehicle. This test was very slow to start, it was difficult to integrate the full vehicle, the state of the hardware was immature, and coordination between different groups was quite weak. It still taught the team a lot about the vehicle and provided a good understanding of which systems were easy to integrate and which were not, but the test was aborted after damaging a harness on final assembly, and operations never began. Competition simulator has high potential for being an effective tool to gauge preparedness, and future years will teach the team more about how to run tests like these.

In general, integration testing in 2024 was very successful and revealing for which systems were prepared for flight and which required more development. One critical success was testing as soon as systems were ready, regardless of whether it was a “complete” test (i.e. filling the vehicle with nitrogen gas and testing valve actuation even without the complete electronics to support it, which did reveal issues in valve design). One key failure was testing pace being too quick, not allowing for issues from one test to be resolved before beginning the next.

Risk Analysis

Risk analysis on Waterloo Rocketry is a very informal process, mostly driven by the instincts of system experts. In general, system experts will uncover, understand, and then mitigate, eliminate, or accept any risks that are described. Additionally, risk analysis is undertaken for each report, and discussion of interface risks happens informally frequently. Risky (propulsion) or poorly understood (airbrakes) systems will get extra risk attention, and novel risks are free to be brought up to the relevant experts at any time.

Informal systems such as these leave many risks uncovered, especially those that are low likelihood. It seems unlikely that any more formal risk gathering process would reveal any more issues than regular, informal conversation would.

The team also does not have any system for tracking risks or ensuring that they are appropriately addressed, this responsibility is typically left to the relevant expert, with stakeholders occasionally checking in. The lack of systems or effective processes for risk followup is a major issue that is guaranteed to cause interface and flight problems at some point, even if none were seen during the development and flight of *Borealis*.

Pre-flight Operations, Flight, and In-flight Failures

With respect to the mechanical and integration systems of *Borealis*, flight performed nominally. However, pre-flight operations were challenging, with many mistakes being made early in design being punishing during final assembly and operations. For example, the high number of joints on the vehicle made assembly quite slow. Additionally, the wide spread of electronics and power sources forced much of the vehicle to be assembled on flight day (L-0), which made flight day very busy. Future vehicles should be simpler to assemble, and the work should be able to be pushed as early as possible to allow for more time between assembly and flight day.

There were three key issues encountered during operations and flight. The first is a pneumatics leak when the fill port disconnected from the vehicle. This failure was accepted. The second was airbrakes' failure to activate, and the third was the parachute's failure to disreef. It has been determined that none of these problems are interface or integration related problems and therefore will not be discussed in this report.

Conclusion and Recommendations

These are the key recommendations from this report, largely targeting process improvements rather than specific technical decisions

- A stronger link should be formed between general rocket requirements and the requirements found in project proposals. This will ensure that no requirements are forgotten and all can be crosschecked.
- Relatedly, any requirements changes must be made in a public space with all relevant parties being notified of the change. This should also be relayed back into the governing document of that project (typically a project proposal).
- Design reviews are an underutilized tool and should be implemented at two different phases of the project, after preliminary design (PDR) and after detailed design (CDR).

- Suggestions should be acquired from the team on how to improve Waterloo Rocketry's Skeleton CAD process to guarantee that it is effective for everyone.
- The integration lead must either better understand electrical systems and development or recognize that it is a gap and request help from the system experts. In future, it could be effective to have a mechanical integration lead and an electrical integration lead that work closely together.
- Waterloo Rocketry has poor practices for followup on key risks. This is a major issue that will cause problems when established risks are forgotten about or their status is unclear. Systems for better risk followup should be implemented and communicated effectively.

In general, system integration on *Borealis* worked effectively and led to in-flight success. However, several key deficiencies have been noted and expanded upon in this document. These practices will help future integration leads to avoid these mistakes and improve the vehicle integration process.

References

- [1] R. I. Faulconbridge and M. J. Ryan, managing Complex Technical Projects: A Systems Engineering Approach, Norwood: Artech House, 2003.
- [2] NASA, NASA Systems Engineering Handbook, Washington, DC: NASA, 2007.
- [3] NASA, What Made Apollo A Success?, Washington, DC: NASA, 1970.